

Industrial Knowledge Intelligence Platform (PS-8)

Technical Architecture & Operational Specification Document | Unified Asset & Operations Brain

Problem Statement	PS 8: AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain
Project Name	ET Industrial AI
Repository URL	https://github.com/Saurabh-Shinde-Patil/ET-industrial-ai
Date	July 20, 2026

1. Executive Summary

In thermal power plants, chemical process facilities, and heavy manufacturing installations, critical operational knowledge is frequently fragmented across thousands of static PDF manuals, handwritten maintenance logs, Standard Operating Procedures (SOPs), and historic Root Cause Analysis (RCA) records. During acute equipment trips or forced outages, plant engineers face significant delays sifting through non-searchable document repositories, leading to extended downtime, elevated safety risks, and operational inefficiencies.

ET Industrial AI addresses this operational bottleneck by deploying a unified, enterprise-grade Knowledge Intelligence Platform. Built on a microservices architecture, the system combines a Zero-Hallucination Retrieval-Augmented Generation (RAG) conversational engine, a high-density vector search index (FAISS), an automated 5-Whys Root Cause Analysis engine, and predictive maintenance risk scoring. The platform provides plant engineers with immediate, contextual answers backed strictly by ground-truth document citations, enforcing strict confidence thresholds to prevent AI hallucination.

2. Problem Statement Alignment (PS-8)

The project is engineered specifically for Problem Statement 8: AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain. The table below outlines how specific plant operational challenges are addressed by the platform architecture:

Industrial Operational Challenge	ET Industrial AI Architectural Solution
Knowledge Isolation across Plant Siloes	Centralized document ingestion & OCR parsing supporting PDF, Word, TXT, and scanned image formats with direct asset tagging.
Risk of AI Hallucinations on Critical Assets	Zero-Hallucination RAG Chain enforcing similarity threshold suppression (<50% score). Returns standardized fallback notices when manuals lack matching context.
Repetitive Equipment Failures	Automated 5-Whys Root Cause Analysis engine generating standardized incident reports and Corrective Action (CAPA) plans.
Unplanned Equipment Downtime	AI Predictive Maintenance engine calculating asset failure risk probabilities based on telemetry parameters and operating hours.
Enterprise Security & Role Compliance	8 RBAC roles (Plant Operator, Maintenance Eng, Reliability Eng, Safety Officer, Production Eng, Plant Mgr, Knowledge Admin, System Admin) with audit event logging.

3. System Architecture & Technical Implementation

The system is structured as a decoupled, multi-tier microservice architecture to ensure high throughput, horizontal scalability, and strict separation between the API gateway, vector store, and AI inference engines.

3.1 Core Components

1. Frontend Web Client (React 18 / Tailwind CSS / Lucide Icons):

Provides a dark-mode interface with operational dashboards, RAG conversation view, document manager, asset hierarchy, and admin audit tools.

2. Node.js Express REST API Gateway (Port 5000):

Handles user authentication, JWT session validation, RBAC route protection, MongoDB persistence, audit event logging, and proxy dispatch to the Python AI service.

3. Python FastAPI AI Microservice (Port 8000):

Houses core AI engines: SentenceTransformers embedding generation, FAISS vector indexing, Reciprocal Rank Fusion (RRF) hybrid search, PyPDF2/pdfplumber text extraction, Tesseract OCR, and RAG synthesis.

4. Detailed AI & Vector Search Pipeline

4.1 Document Processing & Vector Chunking

When an engineer uploads an industrial document, the system executes a three-phase pipeline:

- 1. **Text Extraction:** Extracted using pdfplumber or Tesseract OCR (for scanned engineering drawings).
- 2. **Chunking:** Text is segmented into 500-token sliding windows with a 100-token overlap to preserve continuity.
- 3. **Dense Vector Embedding:** Encoded into a 384-dimensional dense vector using SentenceTransformers ('all-MiniLM-L6-v2'). Indexed into FAISS (IndexFlatIP) for cosine similarity search.

4.2 Zero-Hallucination RAG Chain & Guardrails

To guarantee operational safety on live plant assets, the engine applies similarity score thresholds:

- **Score $\geq$ 80%:** High Confidence answer generated with ground-truth citations.
- **Score 50% – 79%:** Medium Confidence answer synthesized.
- **Score $<$ 50%:** Generation is suppressed. The AI returns a standardized notice stating that no matching context exists, avoiding hallucinated instructions.

5. Plant Verification & Benchmark Test Results

Asset & Document	Test Query Statement	Retrieved Context & Citation	Result Status
PUMP-101 (SOP)	What is the startup sequence and seal pressure?	SOP-PUMP-101 v2.1: Plan 53A pressure at 4.2 bar, startup MOV-102 over 90s.	PASSED (75% Conf)
COMP-07 (RCA)	What was the root cause of bearing temp trip?	RCA-COMP-07 v1.2: Debris blockage in oil filter bypass valve causing lube starvation.	PASSED (75% Conf)
Nuclear Core-9	What is core temp limit of Reactor Core 9?	Out-of-Scope query suppressed by safety guardrail (<50% score). No hallucination.	PASSED (Suppressed)

6. Developer Setup & Installation

Repository: <https://github.com/Saurabh-Shinde-Patil/ET-industrial-ai>

1. **AI Microservice:** `cd ai_service && python main.py` (Port 8000)
2. **API Gateway:** `cd backend && npm run dev` (Port 5000, Swagger UI at /docs)
3. **Frontend Client:** `cd frontend && npm run dev` (Port 5173)
4. **Docker Production:** `docker-compose up --build -d`

7. Conclusion

ET Industrial AI establishes a robust, production-grade platform for industrial operational intelligence. By linking legacy document siloes with zero-hallucination vector RAG intelligence, the system reduces maintenance resolution time and improves overall equipment effectiveness (OEE).